

Aircraft Performance & Atmosphere Analyzer

Project Report

1. Introduction

The Aircraft Performance & Atmosphere Analyzer is a Java-based console application that calculates basic atmospheric properties and aircraft performance parameters at different altitudes.

2. Objective

To develop a simple computational tool for analyzing ISA atmosphere conditions and important aircraft performance quantities such as Mach number, lift, drag, stall speed and Reynolds number.

3. Technologies Used

Programming Language: Java. Development approach: Object-oriented concepts, mathematical modelling and console-based user interaction.

4. Main Features

- ISA temperature, pressure and density calculation
- Speed of sound and Mach number
- Dynamic pressure
- Lift and drag
- Stall speed
- Reynolds number
- Performance comparison at multiple altitudes

5. Equations Used

Dynamic pressure: $q = \frac{1}{2}\rho V^2$

Lift: $L = qSC_L$

Drag: $D = qSC_D$

Mach number: $M = V/a$

Stall speed: $V_s = \sqrt{(2W/(\rho SC_{L_{max}}))}$

Reynolds number: $Re = \rho VL/\mu$

6. Result

The program successfully provides numerical estimates of atmospheric conditions and aircraft performance for user-defined flight conditions. It also demonstrates how altitude and air density affect aircraft performance.

7. Conclusion

The project provides a simple foundation for understanding the relationship between atmospheric conditions and aircraft performance. It can be extended with graphical visualization, a GUI and more advanced atmosphere and aerodynamic models.

Project Type: Java Application **Domain:** Aerospace / Aircraft Performance